Snowflake Basics

Creating a Database:

```
CREATE DATABASE IF NOT EXISTS ECOMMERCE;
SHOW DATABASES LIKE 'ECOMMERCE';
```

Creating a Schema and Assigning it to the Database:

```
CREATE SCHEMA IF NOT EXISTS RAW;
USE SCHEMA ECOMMERCE.RAW;
```

Creating a Warehouse:

```
CREATE WAREHOUSE IF NOT EXISTS ECOMMERCE_WH
WAREHOUSE_SIZE = 'XSMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE
INITIALLY_SUSPENDED = TRUE;
```

Creating a Table within ECOMMERCE.RAW:

```
CREATE TABLE IF NOT EXISTS SALES (
        ORDER_ID NUMBER(10, 0),
        PRODUCT VARCHAR(100),
        QUANTITY NUMBER(10, 0),
        UNIT_PRICE NUMBER(12, 2),
        CITY VARCHAR(100)
);
```

Creating File Format:

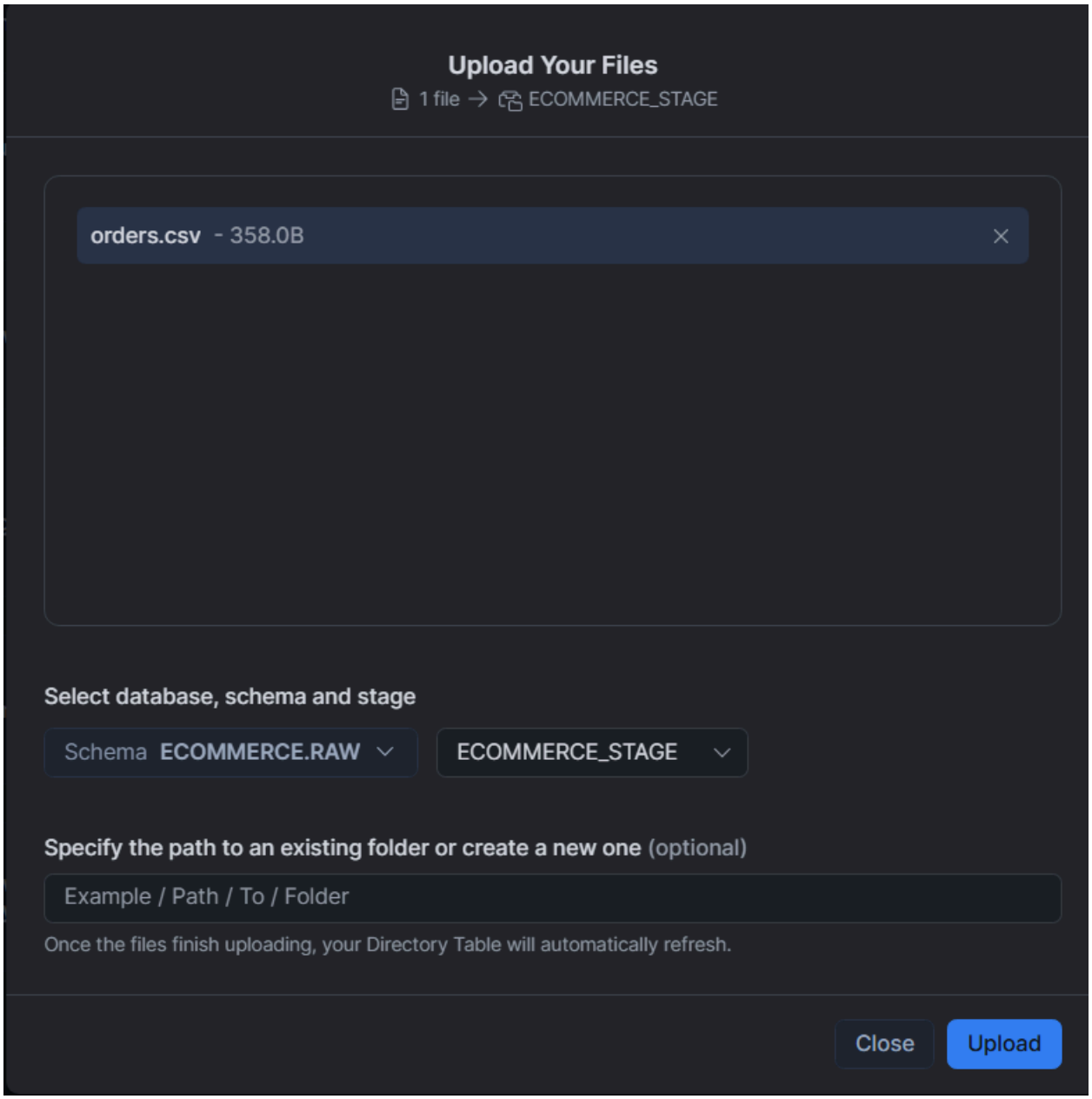
```
CREATE FILE FORMAT IF NOT EXISTS CSV_FORMAT
    TYPE = CSV
    SKIP_HEADER = 1
    FIELD_OPTIONALLY_ENCLOSED_BY = '"'
    EMPTY_FIELD_AS_NULL = TRUE;
```

Creating Stage:

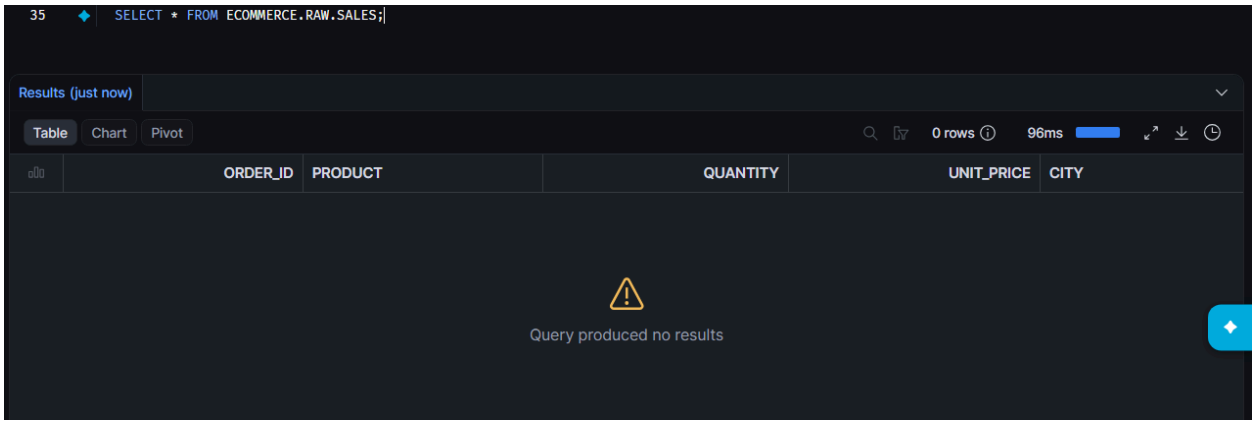
```
CREATE STAGE IF NOT EXISTS ECOMMERCE_STAGE
    FILE_FORMAT = 'CSV_FORMAT';
```

As specified, it uses the file format we created earlier.

Uploading the CSV file into ECOMMERCE.RAW.ECOMMERCE_STAGE:



Verifying Data in table:

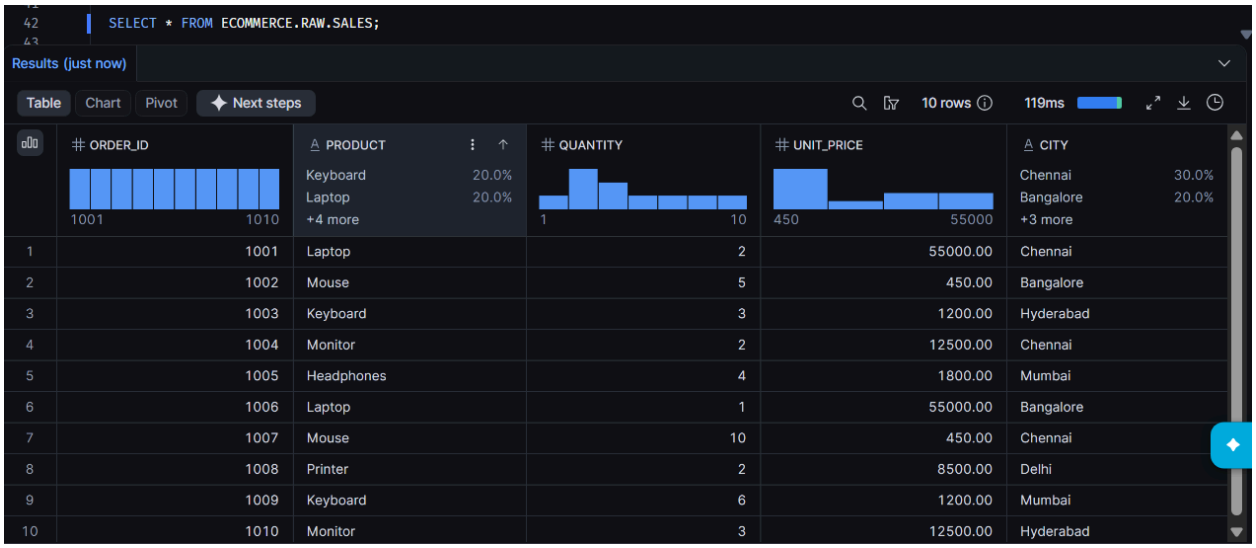


The data is loaded into the stage, ECOMMERCE.RAW.ECOMMERCE_STAGE.
For the data to be reflected in the table, we must perform the COPY INTO operation.

COPY INTO Operation:

```
COPY INTO ECOMMERCE.RAW.SALES
FROM @ECOMMERCE.RAW.ECOMMERCE_STAGE
FILE_FORMAT = 'CSV_FORMAT'
ON_ERROR = 'ABORT_STATEMENT';
```

Verifying Data in Table:



Data has been successfully loaded into the table

Query Operations:

1. Selecting Distinct Products:

```
44  SELECT DISTINCT PRODUCT
45  FROM ECOMMERCE.RAW.SALES;
```

Results (just now)

Table | Chart | Pivot

	PRODUCT
1	Laptop
2	Mouse
3	Keyboard
4	Monitor
5	Headphones
6	Printer

2. Retrieving Orders from a City:

```
47  SELECT *
48  FROM ECOMMERCE.RAW.SALES
49  WHERE CITY = 'Chennai';
```

Results (just now)

Table | Chart | Pivot — 3 rows — 330ms

	ORDER_ID	PRODUCT	QUANTITY	UNIT_PRICE	CITY
1	1001	Laptop	2	55000.00	Chennai
2	1004	Monitor	2	12500.00	Chennai
3	1007	Mouse	10	450.00	Chennai

3. Calculating Total Quantity per Product:

```sql
SELECT PRODUCT, SUM(QUANTITY) AS TOTAL_QUANTITY
FROM ECOMMERCE.RAW.SALES
GROUP BY PRODUCT
ORDER BY TOTAL_QUANTITY DESC;
```

Results (just now) — Table | Chart | Pivot — 6 rows — 431ms

	PRODUCT	TOTAL_QUANTITY
1	Mouse	15
2	Keyboard	9
3	Monitor	5
4	Headphones	4
5	Laptop	3
6	Printer	2

4. Calculating Total Revenue for each Product:

```sql
SELECT
    PRODUCT,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE
FROM ECOMMERCE.RAW.SALES
GROUP BY PRODUCT
ORDER BY TOTAL_REVENUE DESC;
```

Results (just now) — Table | Chart | Pivot — 6 rows — 292ms

	PRODUCT	TOTAL_REVENUE
1	Laptop	165000.00
2	Monitor	62500.00
3	Printer	17000.00
4	Keyboard	10800.00
5	Headphones	7200.00
6	Mouse	6750.00

5. Top 5 Orders based on Quantity:

```sql
SELECT *
FROM ECOMMERCE.RAW.SALES
ORDER BY QUANTITY DESC
LIMIT 5;
```

Results (just now) — Table | Chart | Pivot — 5 rows — 29ms

	ORDER_ID	PRODUCT	QUANTITY	UNIT_PRICE	CITY
1	1007	Mouse	10	450.00	Chennai
2	1009	Keyboard	6	1200.00	Mumbai
3	1002	Mouse	5	450.00	Bangalore
4	1005	Headphones	4	1800.00	Mumbai
5	1003	Keyboard	3	1200.00	Hyderabad